NGHIA H. NGUYEN

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GitHub ⋄ LinkedIn

RESEARCH INTERESTS

Machine Learning, Optimization, Interpretability, Algebraic Structure in LLMs

EDUCATION

University of Pennsylvania, Philadelphia, PA — *2023 – Present*
PhD in Computer Science, advised by René Vidal

University of Science, VNU-HCM, HCM City, Vietnam — *2017 – 2021*
Bachelor of Science in Computer Science

SELECTED WORKS

(* indicates co-first authors.)

- "Hierarchical Concept Embedding & Pursuit for Interpretable Image Classification," **Nghia Nguyen**, Tianjiao Ding, René Vidal, ***CVPR***, 2026. [paper link]
- "SSD: Sparse Semantic Defense against Semantic Adversarial Attacks to Image Classifiers," **Nghia Nguyen**, Darshan Thaker, Konstantinos Emmanouilidis, Tianjiao Ding, René Vidal, ***UCRL Workshop @ ICLR***, 2026.
- "Improving Neural Ordinary Differential Equations with Nesterov's Accelerated Gradient Method," **Nghia H. Nguyen***, Tan M. Nguyen*, Huyen K. Vo, Stanley Osher, Thieu N. Vo, ***NeurIPS***, 2022. [paper link]
- "Towards a Robust WiFi-based Fall Detection with Adversarial Data Augmentation," Tuan-Duy H. Nguyen, **Huu-Nghia H. Nguyen**, ***CISS***, 2020.

RESEARCH EXPERIENCE

FPT Software AI Center, Ho Chi Minh City, Vietnam — *Oct. 2021 – Jul. 2023*
AI Research Resident (Predoctoral Researcher)

- Advisors: Tan M. Nguyen (UCLA, currently NUS), Thieu N. Vo (Ton Duc Thang University, currently University of Bath).
- Derived the continuous adjoint equation for the Nesterov Neural ODE, an important theoretical result for our work.
- Implemented the method and ran experiments of Nesterov Neural ODE on image classification, continuous normalizing flows, and point cloud separation.

VinBrain, Ho Chi Minh City, Vietnam — *Apr. 2021 – Aug. 2021*
Applied Scientist Intern

- Supervisor: Soan T.M. Duong (University of Wollongong).
- Improved the object detection model mAP@0.5 score by over two times from a pretrained model when tested on a custom version of the Open Images v6 dataset.

LSIR, École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland (Remote from Vietnam) *May 2020 – Oct. 2020*
Research Intern

- Supervisor: Chi Thang Duong (EPFL).
- Worked on a Subtree-Reuse-based SQL Query Optimizer using Graph Neural Network and Box Embeddings.
- Performed experiments on the Attribute Discovery dataset to verify the plausibility of a new Multimodal Feature Fusion method using Graph Neural Networks, achieving a test accuracy of 96.86% on this dataset.

Faculty of Information Technology, University of Science, VNU-HCM, Ho Chi Minh City, Vietnam *Sep. 2019 – Aug. 2020*
Research Assistant

- Discovered the correlation between forking patterns, software health, and risks in software repositories.
- Performed inference in the Software Heritage Dataset to extract forking patterns between 50,000 repositories using PostgreSQL and Neo4J.

ACHIEVEMENTS & AWARDS

- Scholar Award; *Advances in Neural Information Processing Systems (NeurIPS)* 2022. Granted full registration fee and accommodation in New Orleans.
- Recommended candidate; The Vietnam Education Foundation (VEF) 2.0 Program 2022. Passed two interview/review rounds with leading Vietnamese academics and U.S. professors.
- Participant; Pre-PhD Summer School by VinUni-Illinois Smart Health Center, VinUniversity 2022 (Top 50/220 applicants).
- Dean's List; Advanced Program in Computer Science, University of Science, VNU-HCM 2018, 2020 (Top 5/92 students).
- Honorable Mention; Vietnam National Olympiad for Undergraduates in Informatics 2019.
- 2nd place; University of Science VNU-HCM Olympiad in Informatics 2019.

OTHER EXPERIENCE

Big-O Coding, Ho Chi Minh City, Vietnam *Jul. 2018 – Dec. 2020*
Teaching Assistant, Problem Setter, Software Engineer

- Explained Algorithms and Data Structure concepts to students weekly and prepared over 20 programming problems.
- Built version 1 of the Front-end Web Application for Big-O Coding's website to host programming problems and contests for up to 1,900 users using ReactJS.
- Reduced page traversing paths for administrators by 50% by implementing a shortcut to problems.

Pathfinder Vietnam, Ho Chi Minh City, Vietnam *Sep. 2018 – May 2020*
Founder, Product Team Leader, Software Engineer

- Led product team of 6 in implementing the Minimum Viable Product using ReactJS and NodeJS.
- Introduced the product's core feature by implementing the programming testing system.

SKILLS

- Programming: Python, C++, JavaScript, Java (prior experience).
- Web: HTML, CSS, ReactJS, Redux.
- Machine Learning: TensorFlow, Neural ODEs, adversarial robustness.
- Databases & Tools: SQL, PostgreSQL, Neo4J.
- English: IELTS Academic 8.5/9.0 (Reading: 9.0, Listening: 9.0, Writing: 7.5, Speaking: 7.5).

SIDE PROJECTS

- **What The Food app** (2018): Implemented a Convolutional Neural Network to identify 101 types of food using TensorFlow; converted the Keras model to a TensorFlow Lite model and deployed it on an Android application using Firebase ML Kit.

EXTRACURRICULAR ACTIVITIES

- Reviewer: NeurIPS, ICLR, CVPR, ICML, ECCV.
- Volunteer: ICLR 2020, ICML 2021.